# Fair AI in Household Energy Prediction

Jabed Hossain | Research contribution summary

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## RESEARCH QUESTION

Could household energy predictions produce unequal errors across demographic and socioeconomic groups when they rely on limited household information? The project examines how household composition, living conditions and occupant behaviour could be incorporated into the study of prediction fairness.

## MY CONTRIBUTION

I drafted the initial study questionnaire for this project in collaboration with UWE Bristol academics. I structured questions covering household composition, demographic and socioeconomic characteristics, housing, appliances and everyday behaviour so that the study could investigate differences between groups. The academic leads refined and finalised the questionnaire.

## QUESTIONNAIRE DESIGN

The drafts cover factors such as household size, age, disability, education, income, housing tenure and routines that affect energy use. The design connects information about people and their circumstances with household and behavioural factors, providing a basis for investigating where prediction errors may fall unevenly.

## PROJECT PLAN AND STATUS

The project brief proposes survey-based data collection, thermodynamic and simulation modelling of energy-use patterns, analysis of relationships with demographic and socioeconomic characteristics, and subsequent development of a fairness-aware prediction model. These are planned project stages. This summary documents my contribution to research design; it does not report a trained model, completed participant recruitment or measured fairness improvements.

## RESEARCH RELEVANCE

This work concerns the design of evidence needed to evaluate an AI system: which variables should be collected, how group-level differences can be investigated, and what can reasonably be concluded from the resulting data.